

NETWORKS AND COMPLEXITY

Solution 10-5

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 10.5: Targeted attack [Lvl. 2]

Consider an ER random graph of N nodes with mean degree z . An attack removes all nodes of degree 0, half the nodes of degree 1, a quarter of the nodes of degree 2 and so on. Compute the size of the network N_a , and the mean excess degree q_a after the attack.

Solution

The attack is described by

$$r_k = p_k / 2^k \quad (1)$$

Substituting the ER degree distribution

$$p_k = \frac{z^k e^{-z}}{k!} \quad (2)$$

we can write the attack generating function as

$$R = \sum r_k x^k \quad (3)$$

$$= \sum \frac{(zx/2)^k e^{-z}}{k!} \quad (4)$$

$$= e^{-z} \sum \frac{(zx/2)^k}{k!} \quad (5)$$

$$= e^{-z} e^{zx/2} = e^{z(x/2-1)} \quad (6)$$

Since we know R we can now compute the removed proportion of nodes

$$r = R(1) = e^{-z/2} \quad (7)$$

and the surviving proportion of nodes

$$c = 1 - r = 1 - e^{-z/2} \quad (8)$$

which gives us the number of nodes after the attack

$$N_a = (1 - e^{-z/2})N. \quad (9)$$

For example, if the network had a mean degree of zero before the attack, the size after the attack is zero, which makes sense.

To find q_a we recall that the mean excess degree of the ER network before the attack is $q = z$. Using a result from the lecture we can compute the reduction of the mean excess degree

by the attack as

$$\delta = \frac{R''(1)}{z} \quad (10)$$

$$= \frac{1}{z} \frac{\partial^2}{\partial x^2} e^{z(x/2-1)} \Big|_{x=1} \quad (11)$$

$$= \frac{1}{z} \frac{\partial}{\partial x} \frac{z}{2} e^{z(x/2-1)} \Big|_{x=1} \quad (12)$$

$$= \frac{1}{z} \frac{z^2}{4} e^{z(x/2-1)} \Big|_{x=1} \quad (13)$$

$$= \frac{z^2}{4} e^{z(x/2-1)} \Big|_{x=1} \quad (14)$$

$$= \frac{z}{4} e^{-z/2} \quad (15)$$

Hence, the mean excess degree after the attack is

$$q_a = q - \delta = z - \frac{z}{4} e^{-z/2} = z(1 - e^{-z/2}/4). \quad (16)$$